

Making Site Search Fast and Relevant

J

Search is an essential feature of most Web sites. Make sure you have search features that are useful and usable. Pay special attention to which words customers type in, how you present the results, how customers interact with the results, and what happens if they can't find what they're looking for. This pattern group deals with designing the interaction so that your customers' searches are more effective.

J1

SEARCH ACTION MODULE

J2

STRAIGHTFORWARD SEARCH FORMS

J3

ORGANIZED SEARCH RESULTS

J1 SEARCH ACTION MODULE



Figure J1.1

IMDb's Web site provides an effective search feature—one that is both simple and powerful.

* BACKGROUND

As a standard element of MULTIPLE WAYS TO NAVIGATE (B1) and a counterpart to BROWSABLE CONTENT (B2), this pattern explains the search half of the search and browse combination.

B1
B2

* PROBLEM

Customers sometimes want to jump quickly from one location to another, but search pages are often too complex for such functionality.

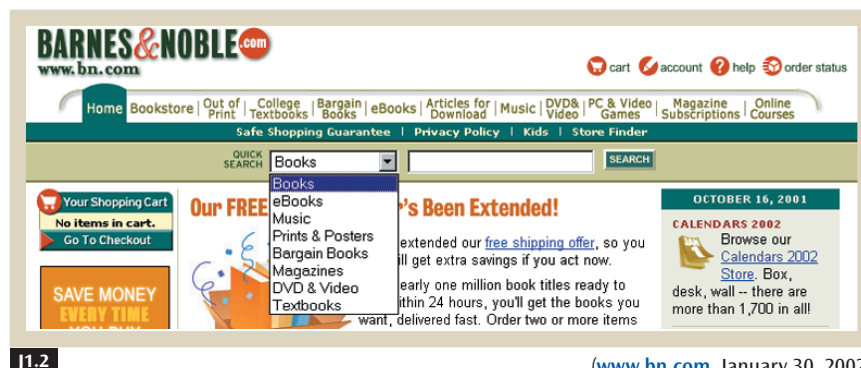
When entering your site, customers sometimes know exactly what they're looking for. A quick and simple search function on the homepage is critical to win these visitors' confidence. Other visitors may browse through a site and then perform a search. If they don't spot a search tool immediately, or they have to go through a complex search page, they are less likely to spend the time. Building a simple search action module into every page is the best way to serve the needs of all customers.

Create a Simple Search Tool • A simple tool for searching, when possible, is much easier to understand than a complicated search page. Although advanced search tools might allow customers to look for words near another word, to search for specific words and not others, and to enter other complex search expressions, most customers would rather have a simpler tool that works well and returns ORGANIZED SEARCH RESULTS (J3). Use simple phrasing to identify the search field and provide an ACTION BUTTON (K4) for starting the search.

Some large Web sites use a search selector to help narrow the focus of a search. Figure J1.2 shows the search selector at Barnes & Noble's Web site. When designing a search selector, keep in mind two important issues. The first is to make sure that the search selector has the right default, in case it goes unnoticed. For example, the default for Barnes & Noble's site is to search on "Books." Anyone who types in something without changing the default will expect to find books on the site of a

Figure J1.2

Barnes & Noble's Web site provides a search selector that lets people search on specific categories, such as "Books" and "Music."



(www.bn.com, January 30, 2002)

J1 SEARCH ACTION MODULE

store known primarily for selling books. The second issue is size. Smaller sites that have fewer than a couple hundred pages in various categories will not be able to yield enough results to warrant a subsection search selector.

We advocate starting simply, using a straightforward search mechanism that lets people type in whatever terms they want. However, when your site becomes large and produces dozens of results to common searches, it's time to add a search selector to let people narrow their searches.

Put the Search Tool in a Consistent Place • People expect to find the search tool in the same place on every page. They might use the search function from any page on your Web site, so place the search action module ABOVE THE FOLD (I2) in the top left, middle, or right, and keep it there on every page.

I2

* SOLUTION

Build a search action module into every page, using simple phrasing that indicates the search space for typing in words or phrases and providing an action button for starting the search. If you have a large site and want to give customers the ability to search a subsection, add a list of subsections and the word *for* to indicate the string to look for.

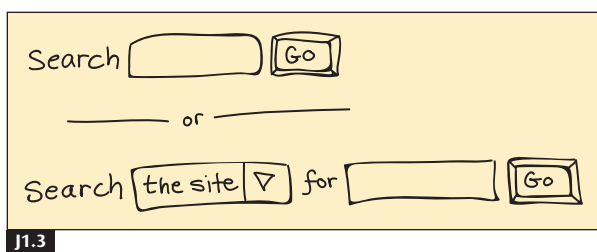


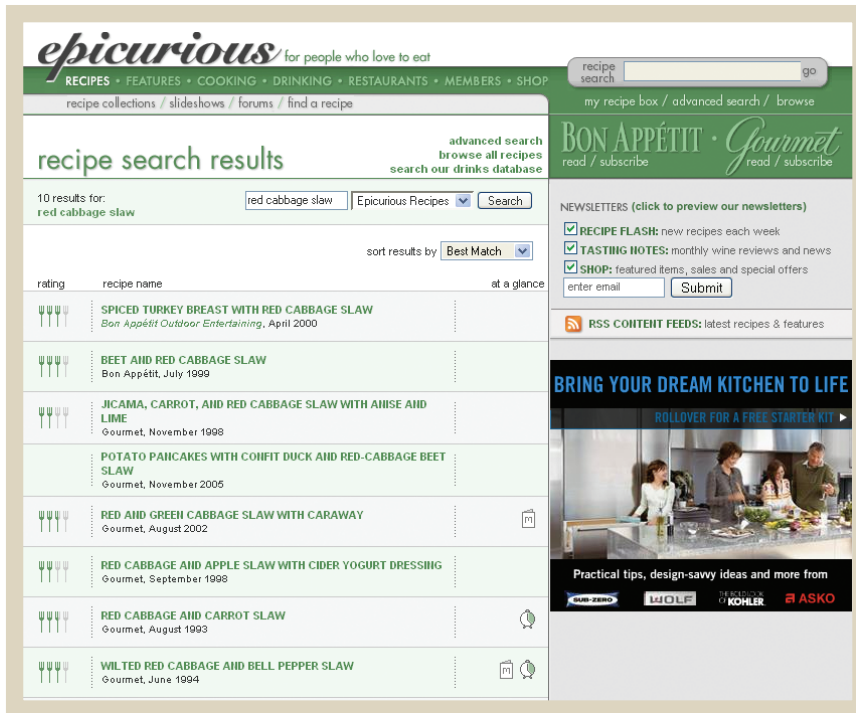
Figure J1.3

Customers will readily use the search action module if it is simple and appears on every page.

* OTHER PATTERNS TO CONSIDER

- I2** Keep your search tool in a consistent place on every page, ABOVE THE FOLD (I2) at the top left, middle, or right.
- D6** If you're WRITING FOR SEARCH ENGINES (D6) when you develop your site's search facility, customers will use your site's content more readily because the search action module will exist on every page.
Make the button that starts the search, which is often labeled "Search" or "Go," an ACTION BUTTON (K4).
- K4** When necessary, customers can fill out more detailed STRAIGHTFORWARD
- J2** SEARCH FORMS (J2). Results will always be returned on an ORGANIZED
- J3** SEARCH RESULTS (J3) page.

J2 STRAIGHTFORWARD SEARCH FORMS



J2.1

(www.epicurious.com, June 16, 2006)

Figure J2.1

Epicurious's search form does not require much precision from the customer. Searching for "red cabbage slaw" results in recipes that have the exact term *red cabbage slaw* near the top, with variants like *red cabbage* and *carrot slaw* included as well.

* BACKGROUND

SEARCH ACTION MODULES (J1) make it easy for your customers to find the search feature on your site. However, the way search terms are specified can confuse customers. This pattern describes how to avoid common errors with search entry forms.

J1

* PROBLEM

Search forms are often ineffective because they require too much precision.

Search is an essential feature for most Web sites. However, most search facilities have limited value because they're designed to make searches easy for the computer to process, instead of being useful and usable for customers.

Boolean Expressions Are Hard to Understand • Boolean searches use the terms *and* and *or*. For example, the search "computer and monitor" would find all Web pages containing both *computer* and *monitor*, whereas "computer or monitor" would find all Web pages containing at least one of *computer* or *monitor*.

It's easy to add Boolean searches to a search engine. However, many studies have shown that customers find Boolean searches difficult to understand. The basic problem stems from how we use *and* and *or* in everyday conversation. For example, whereas a person would say, "I want to search for information about cats *and* dogs," the corresponding search term for computers is actually "cats *or* dogs." Help your customers by providing a search engine that does not use Boolean operators at all, or by providing a search form that has explicit phrasing, such as Google's "with *all* of the words" and "with *any* of the words."

Exact Matches Mean No Matches • Some search engines require exact matches. For example, a customer searching for "presentation" might not get any results if she types in the plural "presentations" or misspells it as "presentatino." Getting the response "no matches were found" may make customers think that there are no results at all. Web search engines can help compensate by using a dictionary to check for plurals and for misspellings.

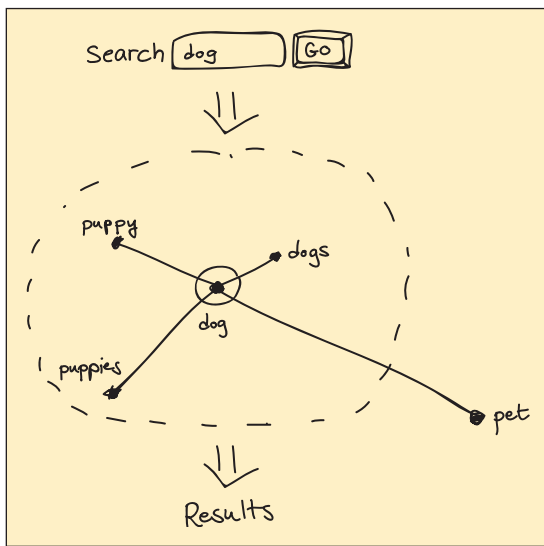
The problem of exact matches also crops up with product names. For example, is it a *laptop* or a *notebook* computer? Is it *apparel* or *clothes*? Your customers will use different terms to mean the same thing, and the development team should design the search engine to take synonyms into consideration. Analyzing your log files, as described in ORGANIZED SEARCH RESULTS (J3), is one way to do this.

Provide Defaults for Category Searches • Some search forms require visitors to specify a category for each search. For example, a music Web site might let people search on categories such as "Artist Name" or "Album Title." But it's easy for customers to end up with no matches because they overlook the category field. To ensure that your customers will get some results, set the default action to be a search on all categories.

J2 STRAIGHTFORWARD SEARCH FORMS

* SOLUTION

Use a search engine that does not require extreme precision. Avoid Boolean searches and exact matches. Compensate for the use of different terms to mean the same thing. Set the search engine to look across all categories by default.



J2.2

Figure J2.2

Search results should include words and concepts that are closely related.

* OTHER PATTERNS TO CONSIDER

Straightforward search forms need to be part of a visible SEARCH ACTION MODULE (J1) that visitors can easily find on a Web page. To assist your customers with searching, it's important to implement CLEAR FORMS (H10). In addition, ORGANIZED SEARCH RESULTS (J3) will make it easier for your customers to understand what the search engine has found for them.

J1

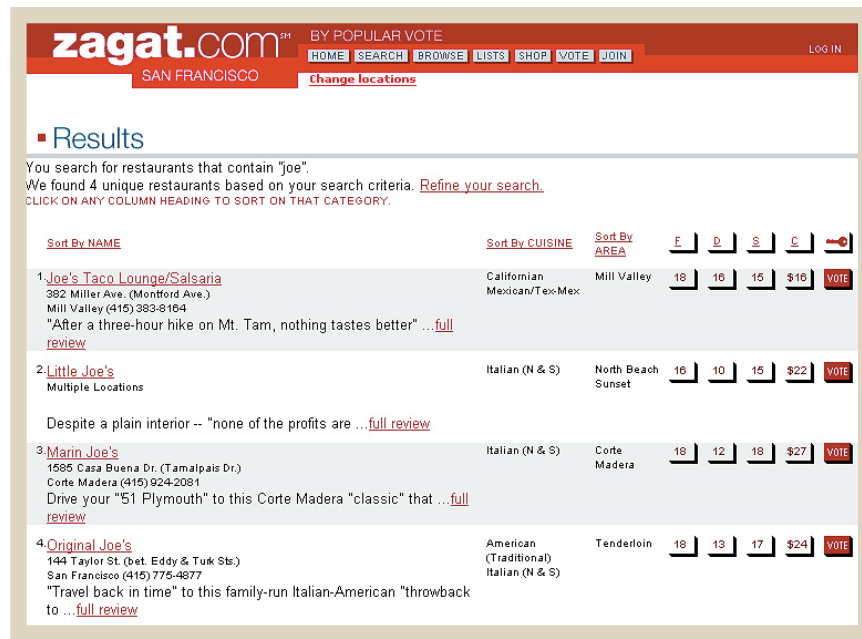
H10

J3

ORGANIZED SEARCH RESULTS J3

Figure J3.1

In its extensive online restaurant guide, Zagat lets its customers sort by name, cuisine, area, food (F), décor (D), service (S), and cost (C). It also shows a restaurant's address and phone number, as well as a short excerpt from the review.



J3.1

(www.zagat.com, August 21, 2001)

* BACKGROUND

J1 J2
H10

A visible SEARCH ACTION MODULE (J1), a STRAIGHTFORWARD SEARCH FORM (J2), and CLEAR FORMS (H10) address some of the common problems that site visitors have using search engines. However, it might still be difficult for your customers to comprehend search results, especially when there are a large number of them. Structuring and organizing search results can make them much easier to understand. This pattern covers ways of arranging and categorizing search results to make them more valuable to your customers. It applies to both local searches on your site and Web-wide search engines.

* PROBLEM

It can be difficult for site visitors to understand search results if there are too few or too many results.

A search engine is only as good as the results it presents. It doesn't matter if it's fast and can simultaneously support thousands of queries per second if people can't understand the results.

Provide Relevant Summaries with the Search Results • The Zagat restaurant guides are indispensable for many city dwellers, providing an overview of all the major restaurants in a metropolitan area. Zagat provides a high-quality online version of its popular restaurant guidebooks. Figure J3.1 shows how zagat.com provides useful, domain-specific information in its search results, including a short excerpt from the full review, as well as ratings of the food, décor, service, and cost of the restaurant.

There are no hard rules here, though. The specific needs of each individual Web site will dictate what should be shown in the search results. On e-commerce sites, for example, it makes sense to show the price and availability of products. And search results on community message boards could show the author and date of posting.

Clearly Organize the Search Results • No one likes sifting through hundreds of hits to find the right one. One way of addressing this problem is to group the hits according to a coherent, logical scheme, such as alphabetically or chronologically. For example, Figure J3.2 shows how Amazon.com groups related search results. This is one way to make sure your search results present customers with **BROWSABLE CONTENT (B2)**.

Provide Good Hyperlinked Titles for Each Hit • Search engines typically display a Web page's HTML title as the name of the search result. Because these titles are what visitors see, it is crucial to give your Web pages **DISTINCTIVE HTML TITLES (D9)**.

Use Log Files to Tailor Results for the Most Common Search Terms • Search engines by themselves don't always provide the best results. The key to better results for your customers lies in your analysis of the log files, which keep track of the terms your customers use to search.

Find the most common terms, and make sure that search results using those terms point to the right place. This approach solves two problems. First, it compensates for the fact that customers use different words to

Figure J3.2

Amazon.com groups search results by category. On a search for “addison wesley computer,” the various hits are grouped by “Books” and “zShops.”



mean the same thing. Figure J3.3 shows how IBM does this on its Web site. The search terms “notebook” and “laptop” point to the same page, greatly increasing the chance that a customer will find the right page.

Second, this approach lets you create the best possible hit for search terms. Look at Figure J3.3 again, and you’ll notice a special area for tailored search results. Whenever a customer searches on “notebook,” or “laptop,” the search page shows a special promotion for selling these computers. However, the regular searches on the site show only ordinary search results. IBM keeps these ordinary search results in case their customers’ “best” hits are not what they’re looking for.

Compensate for Common Misspellings • Figure J3.4 shows how Amazon.com handles misspelled search terms. At the top of the Web page, it notes that there are no search results for the exact search terms entered, but then it corrects the spelling and searches on the new terms. Many customers will not notice that they misspelled the terms, nor will they see the message at the top of the Web page stating that they misspelled the terms, but Amazon.com does the right thing and shows them the results anyway! This is a customer-savvy, elegant design.

Compensating for spelling mistakes means you have to tailor the spell checker’s dictionary to your specific domain. An ordinary spell checker’s dictionary will not be useful for Web sites dealing with legal matters, for example, because it will not contain legal terms.

J3 ORGANIZED SEARCH RESULTS

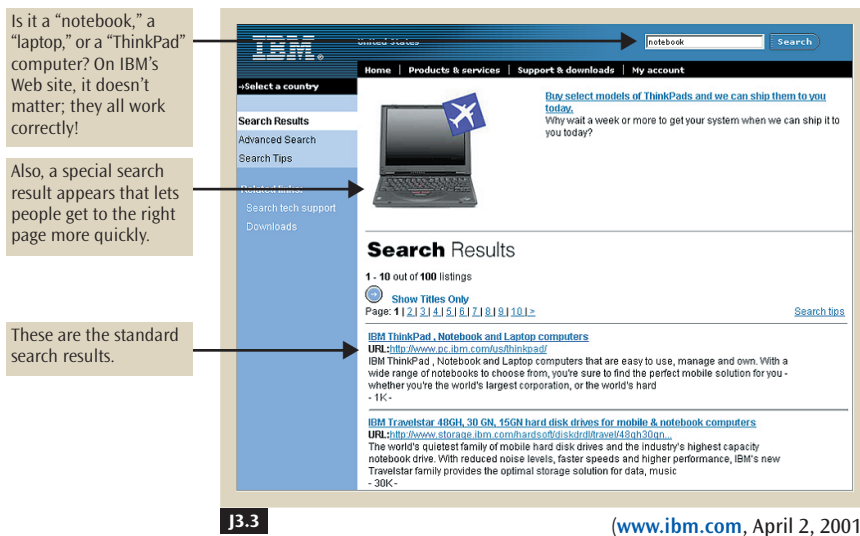


Figure J3.3

IBM presents specially tailored search results for common searches, such as "notebook" and "laptop," in addition to the standard search results.



Figure J3.4

Amazon.com automatically corrects certain misspellings. In this case, it corrects the search "crossing cashm" to read "crossing chasm."

Another way to compensate is to scan your Web server log files for the most commonly misspelled search terms. You will have to tweak your search engine so that these misspellings point to the right terms. This is clearly a tedious proposition, especially if there are a huge number of search terms. Concentrate on finding the most popular terms, perhaps by limiting your corrections to the top one hundred or five hundred.

Provide Support for Common Search Tasks • Common search tasks include things like going to the next page of results, starting a new search, and refining a current search. Figure J3.5 shows a good example of how the Google search engine handles these tasks. The top of the page provides a search entry form, letting visitors quickly create a new search if they did not

Figure J3.5

Google has a clean, minimalist design for displaying search results and supporting common tasks.

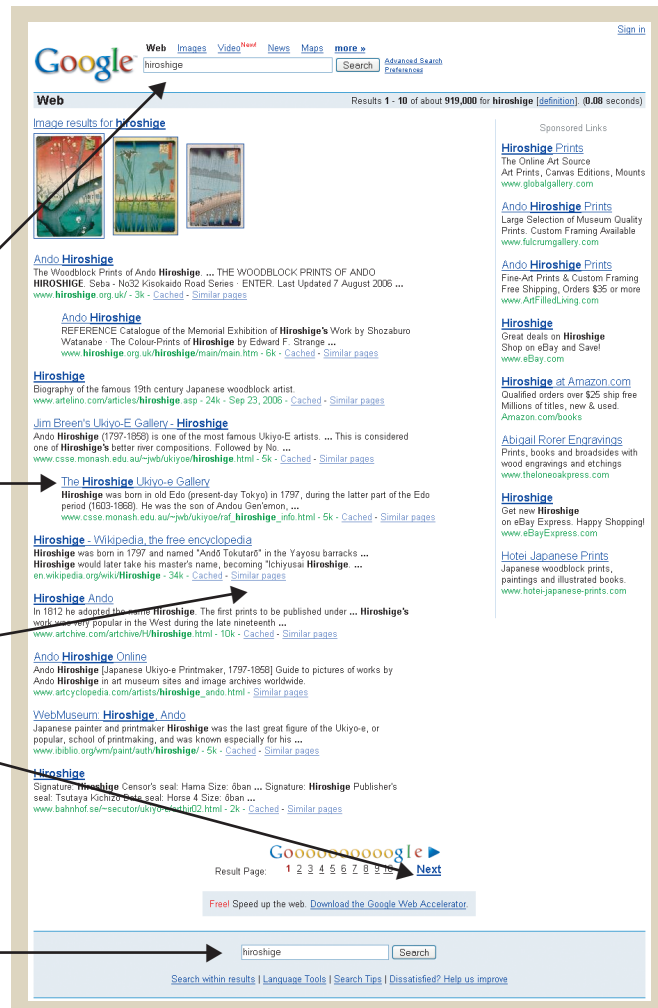
A search area is provided at the top of the page so that people can initiate a new search if the first few search results don't look right.

Results on the same Web site are grouped together so that they don't clutter the page.

People can find more images like this one by clicking on **Similar** pages.

A link (underlined to make it easier to see) takes you to the next page. Note that the image to click on is quite large, making it easy to hit.

A search area is provided at the bottom so that people can search again after skimming the page.



J3.5

(www.google.com, April 15, 2001)

get the results they wanted. In addition, the new-search form includes the search terms that the customer originally typed in, making it easy for them to remember what they entered and to edit it if necessary.

The bottom of the page has a second search entry form because the customer may look through all of the search results for a page, reach the bottom, and then decide to start a new search or refine the existing one. This design is more considerate than forcing the customer to scroll back to the top of the page.

J3 ORGANIZED SEARCH RESULTS

The bottom of the page also contains links to the next set of search results. One subtle but very useful thing that Google's designers have done here is to make the "gle" part above the **Next** button part of the link to the next page of search results. Therefore, the target that customers must click on to go to the next page of search results is fairly large. Incidentally, designs with large targets are also good for improving SITE ACCESSIBILITY (B9) because they help customers with poor motor control, as well as those on a Mobile Web platform.

At first glance, the controls at the bottom might seem to argue against designing for ABOVE THE FOLD (I2). But visitors tend to read the page from top to bottom. If they have not found what they want by the time they reach the bottom, they can go to the next page of results.

* SOLUTION

Provide relevant summaries in your customers' search results. Clearly organize the results. Provide hyperlinked titles for each hit on the search results page. Use log files to tailor the search engine for the most common search terms and to compensate for common misspellings. Support your customers' common search tasks.

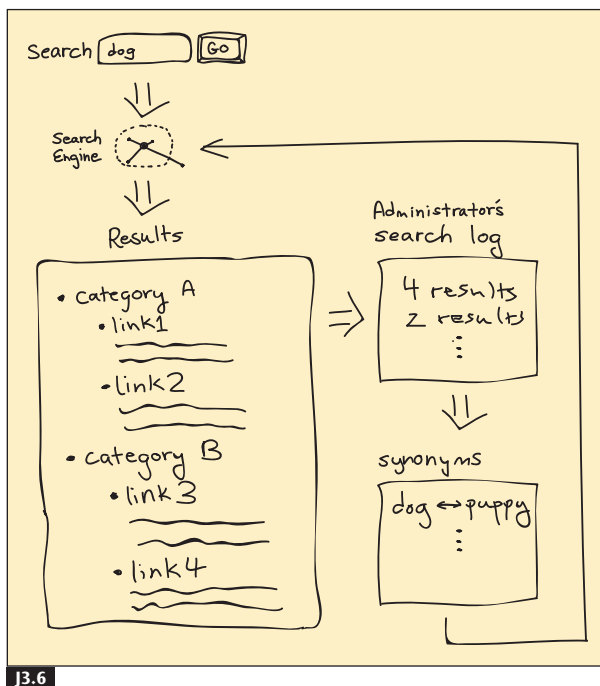


Figure J3.6

Organize search results, and continue to update the search database with common synonyms.

J3.6

* OTHER PATTERNS TO CONSIDER

- 3** You can use the card-sorting techniques described in Chapter 3 (Knowing Your Customers: Principles and Techniques) to find common search terms, but do this before launching the site and before performing a log file analysis. This strategy can help you prevent errors, instead of waiting for them to happen and then fixing the problems after the fact.
- B9** Use the SITE ACCESSIBILITY (B9) pattern to ensure that the results you return can be used by all of your potential customers, including those with disabilities.
- D6** When WRITING FOR SEARCH ENGINES (D6), give your Web pages DISTINCTIVE
D9 HTML TITLES (D9), and construct the pages using simple HTML that a search engine's Web crawlers can understand.
- M3** On mobile platforms, consider using LOCATION-BASED SERVICES (M3) to emphasize nearby places in the search results.

